MINING DEALER BEST PRACTICE

Transmission Carrier Disassembly Tool

Maintenance and Repair | Application | Component Life Management | **Component Renewal (CRC)** | MARC Management

November 2023
4.03-231124-1

1.0 Introduction

One of the key steps in component repairs, is the disassembly process, utilizing the correct tools, to prevent any personal injuries, including any component damage and when certain tools may not be available, the CRC Continuous Improvement Team (CIT) will analyze options, to improve Disassembly and Assembly (D&A) Processes.

This transmission carrier (rotating stand) is one example of the professional approach, the CIT is responsible for at the CRC, to improve workmanship quality and safety.

2.0 Best Practice Description

During our CRC assessment, of the transmission carrier disassembly process, it did offer several opportunities for improvement (i.e).

- Disassembly was performed on a flat bench surface, which prevented removal of the planetary gear shaft(s) resulting in the technician physically sliding the carrier over the edge of the work bench to remove the shaft, creating unsupported sections of the carrier.
- Turning the carrier on the bench surface, would create deterioration of the carrier's face(s) that was in contact with the workbench.
- Large transmission carriers require more effort by the technician to rotate the carrier on the bench surface in order to remove the shaft(s), resulting in potential hand slippage and finger cuts on sharp edges of the carrier.

Based on these observations, it was decided to design a tool that would support the various transmission planetary carriers on a rotary surface, allowing effortless carrier rotation by the technician to remove the shaft(s) from the carrier.

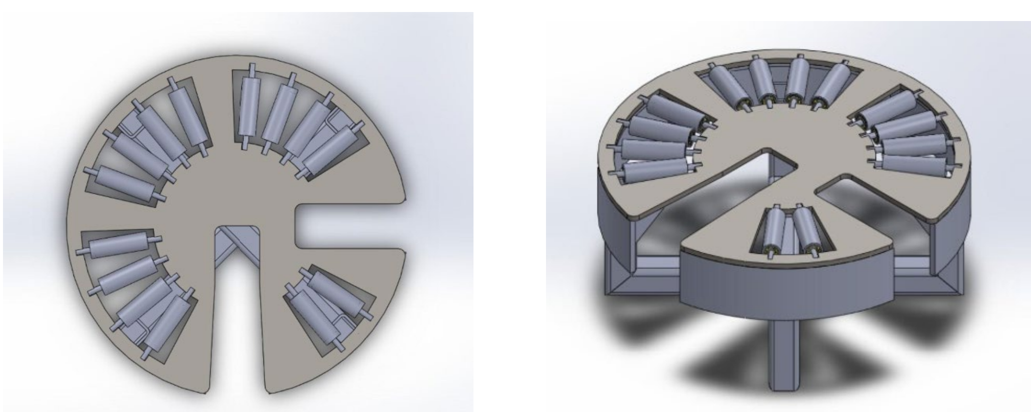

Imagen 1. 3D Modeling.

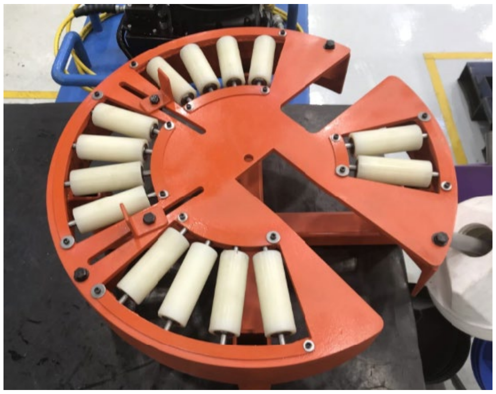

Imagen 2. Upper view.

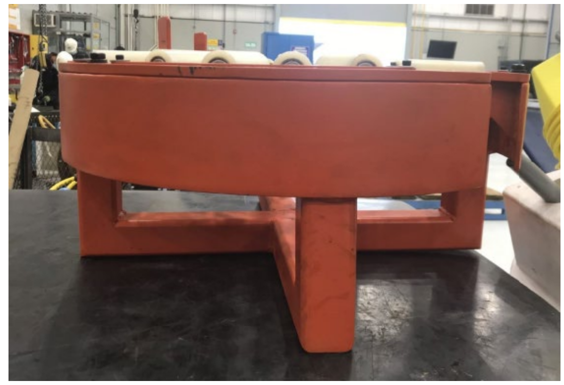

Imagen 3. Side view.

Imagen 4. Bottom view.

Imagen 5. Tool being used.

Imagen 6. Carrier stoppers.

Version 1.2:

1. Multiple rollers of two different lengths, with durable plastic sleeves and sealed bearings were incorporated in the design to support varous models of transmission carriers.
2. The two (adjustable) slide stops were added, for central (carrier) positioning on the tool to eliminate any carrier movement on the rollers, while driving the shaft lock pins out of the carrier bore into the shaft for shaft removal, in either sectioned area of the rotary base.
3. A protecting shelf was added in order to receive the shaft and avoid possible damage.

Image 7. Extended Length Rollers.

Image 8. Clamping levers for stoppers.

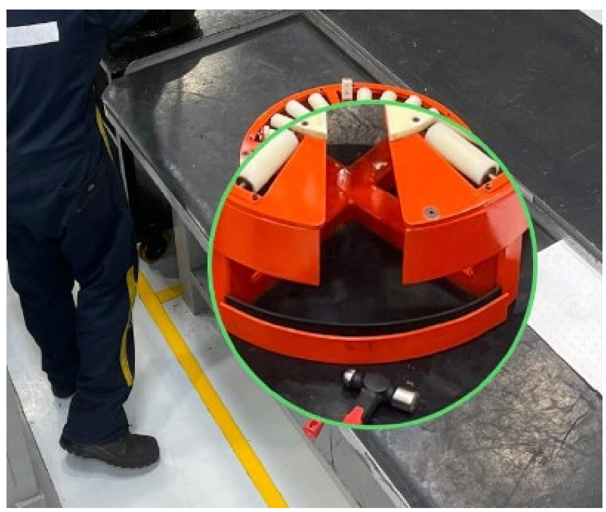

Image 9. Shaft protecting shelf.

3.0 Implementation Steps

Once the design and fabrication of the tooling was completed, the next step was the introduction and commissioning of the tooling to the various CRC technicians assigned to the transmission disassembly workstations and to provide them with proper training in the use of the tool.

The main purpose of this training was to focus on the Central Positioning of the carrier on the rollers, the two quick slide lock adjusters to prevent carrier movement across the rollers when driving out the lock pins from the carrier bore into the shaft and the removal of the shaft from the carrier, in either slot / sectioned area of the rotary base.

4.0 Benefits

The use of this tool will provide additional benefits to the disassembly process of transmission carriers.

- Technician Safety: With the carrier completely supported on a rotary base, will reduce the risk of any mishandling, or injury, as the technician will not need to perform any unsupported movement of the carrier, on the workbench edge, to remove the planetary shaft(s).

- Improve Quality: During the disassembly process, carriers will no longer be slid across or rotated on any workbench surface, eliminating potential carrier surface damage and technician fatigue.

5.0 Resources Required

Parts List:
The tool consists of two main parts, the top plate and the base support.
The top plate holds the rollers on which the carrier will rotate, as well as the slots for the planetary shafts to go through. The base basically supports this plate.

Top plate:
Plate with slots for rollers and shafts to pass through
Rollers (made of nylamid)
Stopper
Eye bolt thread

Base support:
Top plate collar to support the top plate.
Base support for the whole structure, this gives the necessary height for the shaft to come out.
Shaft receiving plate (Shaft will be dropped onto this plate; it must be rubber covered.
Total production cost was about $750.00 USD, this includes material and labor.

6.0 Supporting Attachments / References

- Check attachment for Drawings and Video
- For any further questions, please contact the dealer representative (see below)

7.0 Related Best Practices

N/A

8.0 Acknowledgements

This Best Practice was written by:
Jesús Cabanillas
Asesor Técnico CRC
Empresas MATCO SA de CV
Correo: jcabanillas@matco.com.mx

Recognition:
José Candelario García
Supervisor de desarmado de componentes del Tren de Fuerza
Empresas MATCO SA de CV
Correo: jcagarcia@matco.com.mx